Solution Architecture

Date	02 July 2026
Team ID	SWTID-2026-6128
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

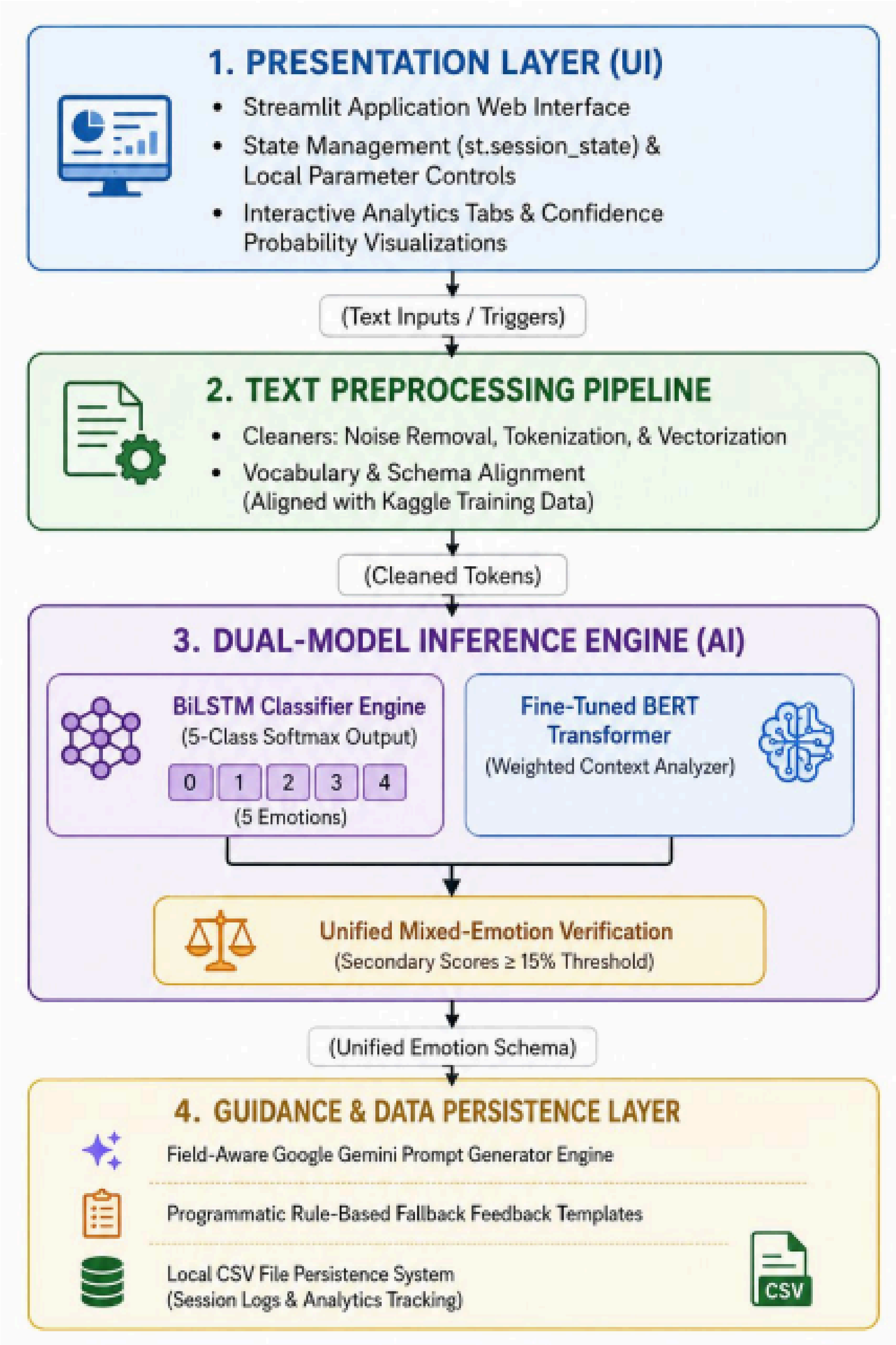
1. Architectural Overview

The platform implements a decoupled, event-driven architecture designed to process user input text, infer psychological/emotional states, and deliver dynamic generative learning interventions. The system isolates the user presentation layer from heavy machine learning inference workloads, maintaining localized caching parameters to meet tight performance and response thresholds ($<500\text{ms}$).

2. Component-Level System Architecture

The solution architecture consists of four primary, interconnected layers. These layers work together in a synchronized pipeline to capture user data, process complex text patterns, infer underlying emotional states, and deliver context-aware educational guidance in real time. By structuring the platform into distinct, decoupled modules, the system ensures that heavy natural language processing (NLP) models can run in parallel without causing lag or freezing the frontend presentation layout.

Each layer is engineered to handle a specific stage of the learner's session, moving smoothly from initial text entry to data cleanup, AI analysis, and final content generation. The flow and responsibilities of these four core architectural pillars are mapped out in the structural blueprint below:



3. Detailed Layer Breakdown

A. Presentation Layer (Streamlit UI Client)

- **Role**: Acts as the primary student/instructor portal.
- **Execution:** Renders native input elements (problem entry boxes, field dropdown selectors, clear/run triggers) and reads internal state

matrices via st.session_state.

- **Output Visualization:** Utilizes pre-compiled dependencies to display live validation bars and analytical breakdowns via Plotly charts without interrupting operational workflows.

B. Dual-Model Core AI Inference Engine

The system uses a parallel routing pattern to process incoming tokenized data arrays through two independent classifiers for maximum accuracy**:** 1. **BiLSTM Layer**: Processes text sequentially, calculating micro sentiments across a 5-Class Softmax distribution.

2. **BERT Layer**: Evaluates context and keywords using customized domain-adapted weights to maintain structural accuracy on technical or highly nuanced phrases.
3. **Unified Evaluation Component:** Merges model inputs. If secondary emotional classes score $\ge 15\%$, the pipeline tags the entry under a mixed-emotion validation state, establishing an accurate, multi layered profile of student confusion.

C. Guidance, Integration, & Regeneration Layer

- **Generative AI Pipeline:** Combines the user's specific text problem, academic discipline, and evaluated emotion into an automated, field aware context prompt dispatched via secure environment configurations (.env) to the Google Gemini API.
- **Fail-Safe Interventions**: If a network failure or an API timeout occurs, a rule-based engine intercepts the state and generates an empathetic fallback response template to maintain user motivation.

D. Data Persistence & Performance Caching

- **Continuous Analytics Logging:** Appends user query logs, model scores, and baseline timestamps down to localized CSV Persistence Storage files to preserve history for long-term progress audits.
- **Memory Efficiency:** Heavy model architecture weights are loaded into an environment-wide memory cache during initialization, eliminating disk-read blockages and allowing immediate textual evaluation loops.